

Data Science Retreat Roadmap

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Today's Agenda



Welcome & Introductions

Getting to know each other and building our learning community



Roadmap

Week-by-week breakdown of skills, projects, and milestones



Software Setup

Welcome to Data Science Retreat!



About Me

Computational physicist transitioning into Machine Learning



PhD at TU Berlin, Computational Physics

Co-founding an AI startup: inny.io



**Key Interests: Multimodal ML,
World Models, Simulations, Agentic AI**



Connect on LinkedIn

Let's Get to Know Each Other



Your Name



Professional Background

Previous experience, or field you're transitioning from.



Why Data Science?

What drew you to this field and this specific bootcamp program.



Program Expectations

What specific skills or knowledge do you hope to gain here



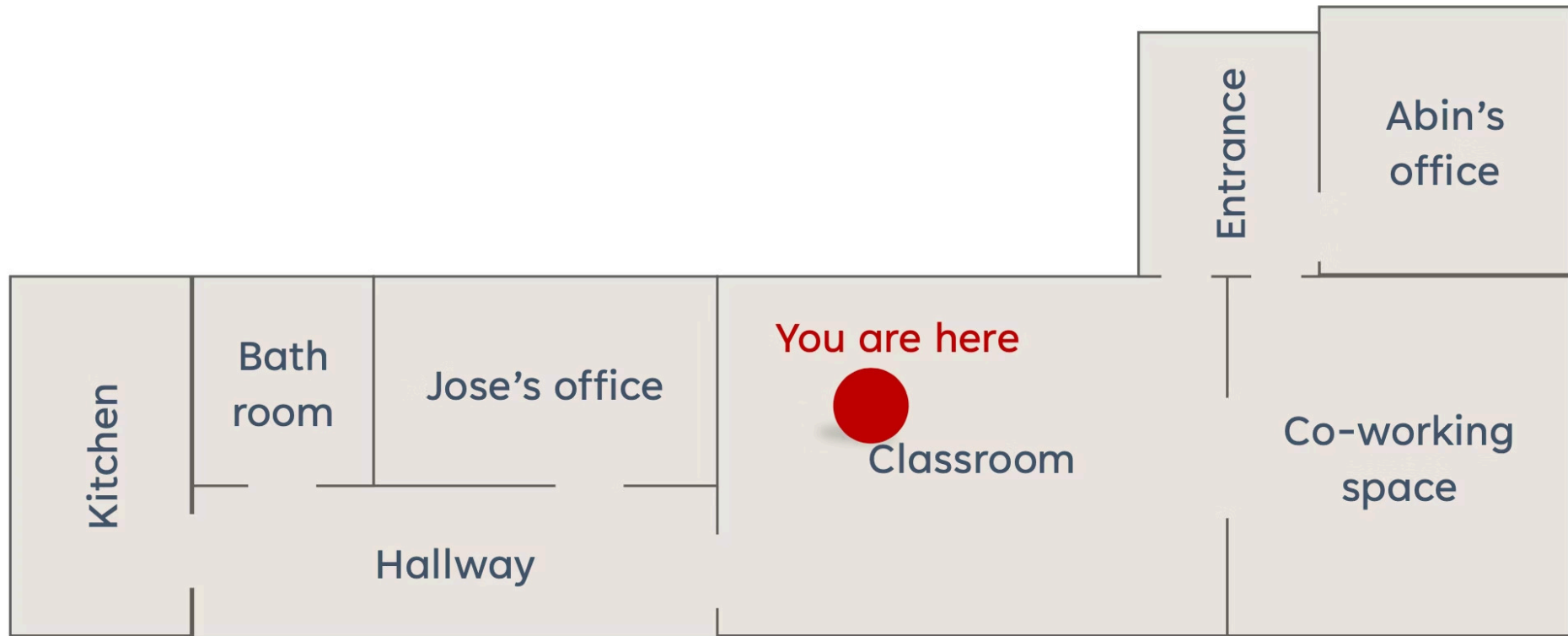
Long-term Goals

Take 2-3 minutes to introduce yourself.

General Info



The Venue



The Teachers



Industry Data Science/ML Experts

Hands-on practical knowledge.
Updated tech stack.



Some are DSR graduates



They are people too :)

Working on their free time to teach at DSR in between their busy schedules.

Organizational Info

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Communication



General Class guidelines

Timing: 09:30 - 17:30
Please be on time.
Be prepared.



Attendance

Participants registered for the in-person format are required to attend physically at the venue.



Feedback

Abin & Arun are always available for any help.

ROADMAP

Programming Fundamentals



Python, numpy, pandas

Introduction to python and the libraries numpy and pandas for Data Science/ML.



Bash/CLI

Introduction to the common line interface and bash commands and actions.



Git

Introduction to git and related workflows.



SQL

SQL for DS.

Statistics Fundamentals

Deep-dive into the statistics and probability topics necessary for Data Science/Machine Learning.

Data Science Fundamentals (Tabular Data)



Data Scraping

Data acquisition, management.



Data Cleaning



Feature Engineering

Modifications and alterations to the data to aid in model development and performance.



Exploratory Data Analysis

Visualizing and summarizing key information and statistics about your data



Model Selection & Evaluation

Deciding on a model that is best suitable for your problem. Evaluating the performance of the model.

Machine Learning Fundamentals (Tabular Data)



Supervised Learning Algorithms

Linear regression, Logistic regression., Support Vector Machines.



Unsupervised Learning Algorithms

K- Nearest Neighbors (KNN), K-Means Clustering, Principal Component Analysis (PCA)



Decision Trees

Decision tree based classification and regression algorithms. Boosting, Bagging etc.

Time Series Analysis



Component Analysis

Trends, Noise, Seasonality.

Goal: *Understand what the data looks like, and what underlying structures exist.*



Time Series Modeling

Autoregressive (AR) processes • Moving Average (MA) processes etc.

Goal: *Fit statistical models to capture dependencies and make forecasts.*



Feature engineering for time series data

Data Visualization



Interactive plots

D3, Plotly



Storytelling & Infographics

Principles of effective visualization

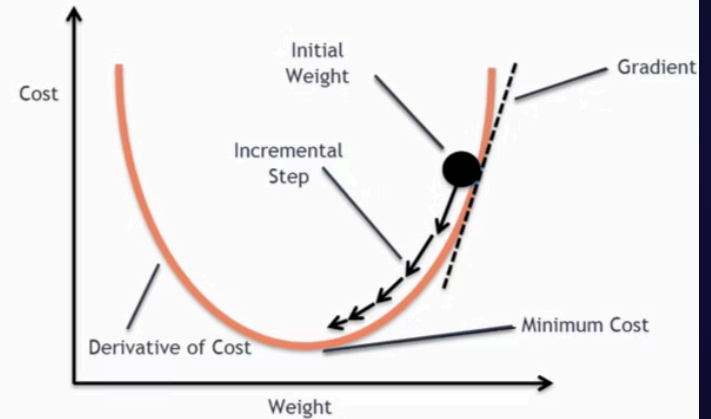
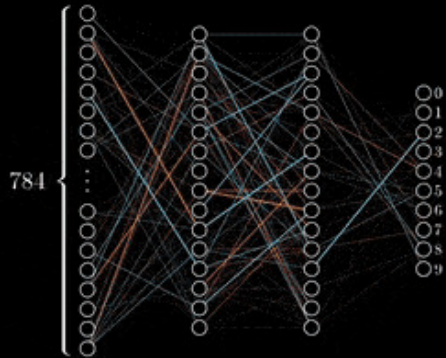
Deep Learning Fundamentals



Fundamentals

Neural Networks, Activation functions, Forward pass, Loss Functions, Back propagation, Gradient Descent, Model evaluation etc.

Training in progress...



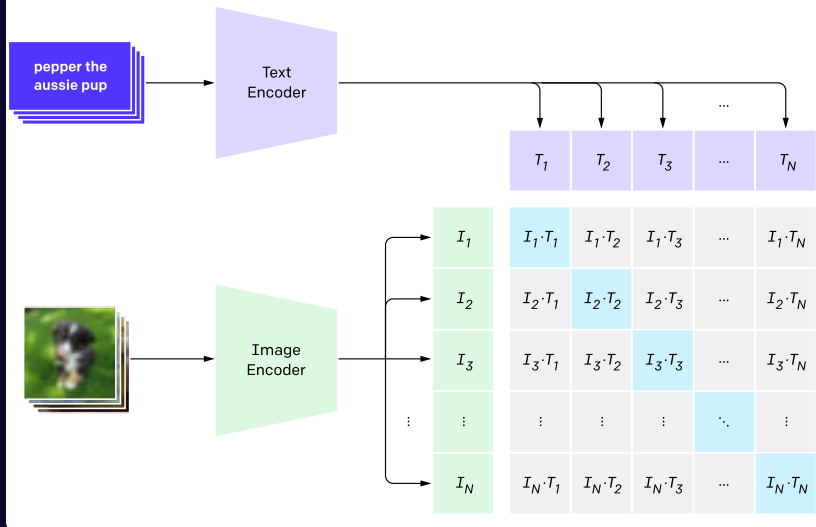
Natural Language Processing



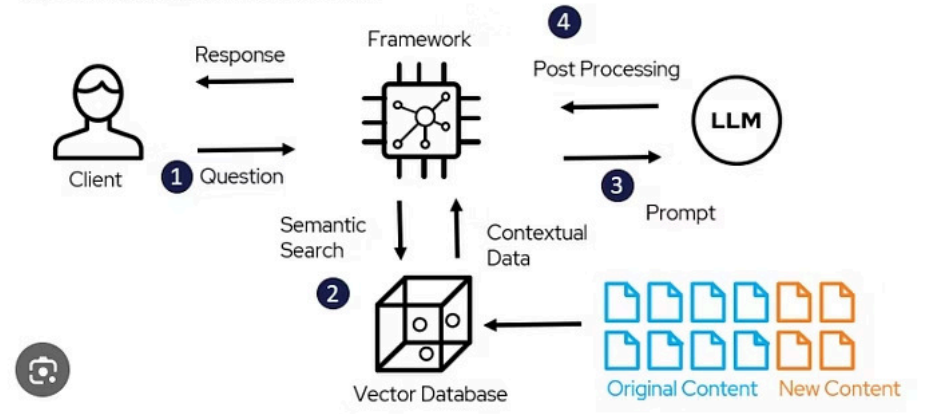
Natural Language Processing

Text embedding algorithms, Text generation (next token predictions), Semantic similarity, Text generation models, Transformers, LLMs, Agentic Workflows, RAG. Fine-tuning, transfer learning, Quantization.

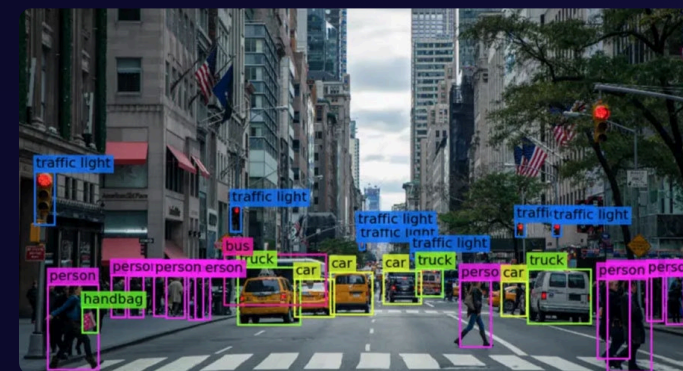
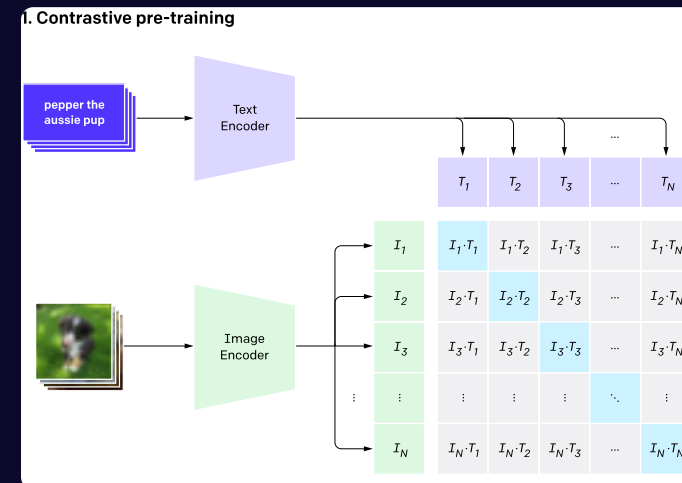
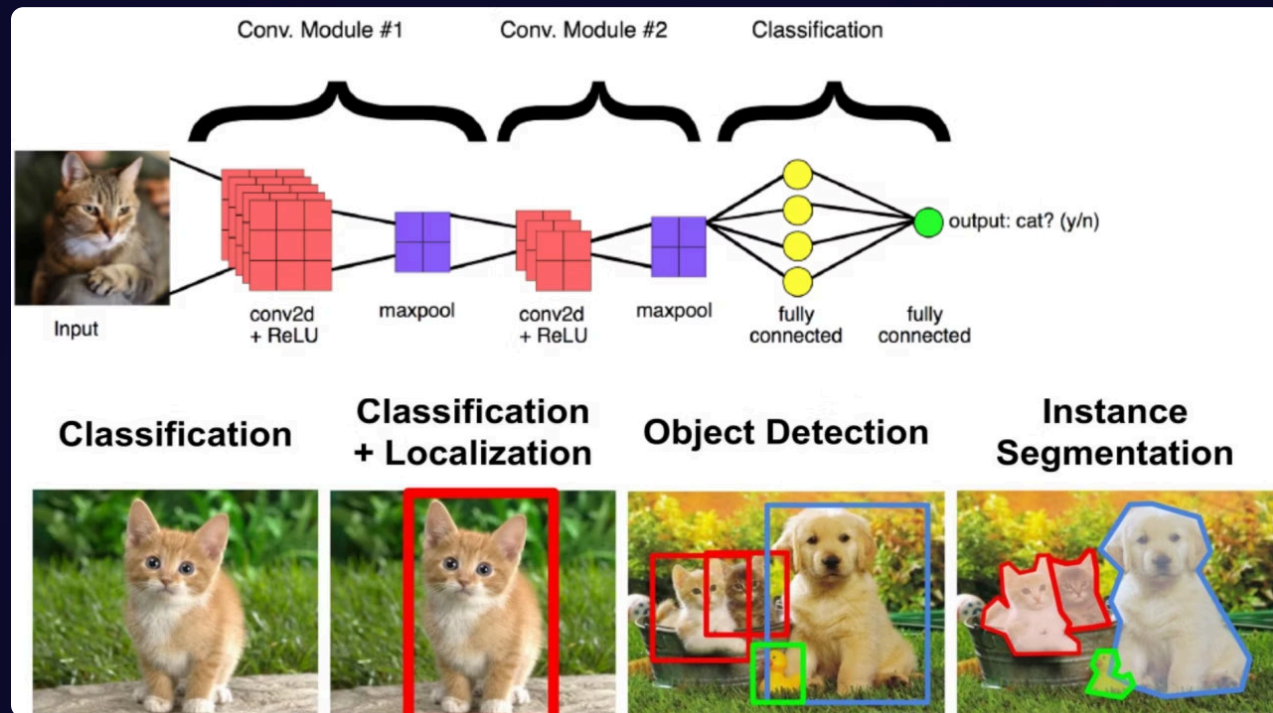
1. Contrastive pre-training



RAG Architecture Model



Computer Vision



Data Science Challenge



Take on a Data competition



Work in teams



Collaboration/Coding best practices

KANBAN boards, Git,



Dates

May 18th & 19th

Agentic AI Workflows

Prompt to Production: Mini Competition

Translate innovative ideas from natural language prompts into fully functional, production-ready solutions in this intensive mini-competition, focusing on real-world application.

Competition Overview



Prompt Engineering

Crafting clear, effective prompts to guide AI model behavior.



Solution Development

Building robust and scalable applications from scratch using coding agents.



Deployment & Demo

Agentic AI



AI Agent Basics

Develop AI agents



Autonomous Capabilities

Explore workflows for reasoning, self-correction, and learning from environmental feedback and outcomes.

Advanced Deep Learning Algorithms



Geometric Deep Learning

Graph Neural Networks (GNNs), Concept of Invariance and Equivariance for GNNs



Reinforcement Learning

Advanced Software Dev Skills (for ML)



Object Oriented
Programming (Python)



Structuring ML projects



Docker



Databases



SQL



Testing

CI/CD pipelines, Types of tests.



UI & Prototype Development in Python

Streamlit



ML Ops

Developing and deploying an ML model. Setting up
automated experiment tracking.

Career Guidance



Session with a Tech Recruiter

1:1 CV Review, insider knowledge about hiring processes.



Business Communication

Techniques for effective communication in interviews and on the job.

Portfolio Project



Mentoring Sessions

Begins on May 13th. The 2 hour sessions with Francisco: Group mentoring sessions. You are free to book further 1:1 sessions with Francisco/Jose after this point.



Decision Day: 23rd June

You should have had sufficient 1:1 meetings and finalized your project by now. Send a 1 page 300 word abstract to Abin/Arun.



Demo Day: July 21st

Final presentation of your project: 10 minutes.

General Tips



LLM usage

Do not overuse it, especially in class, ask your questions to the teachers/peers. Only implement code suggestions that you fully understand.



Session pre-prep

Try to grasp the fundamental concept before each session, so that the tutor can go into the more advanced topics. Go through the Git/Colab notebooks. Install any required software, env setup etc.



Stay Updated

Regularly check all the communication channels: Slack and Google Calendar.



Project Ideas

Start thinking about potential project ideas: the more the better. Do not over-rely on information from LLMs.

Review and Prep Days



Review Day: April 24th, May 13th

Review and ensure that you understand the fundamentals covered until this point.



Prep Day: April 30th

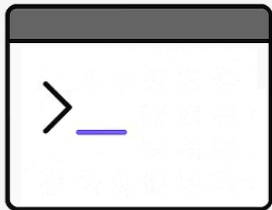
Go through and understand the contents on the repo for the deep learning/ML sessions.

Software Setup

The Command Line Interface (CLI)

CLI

Command-Line Interface



GUI

Graphical User Interface

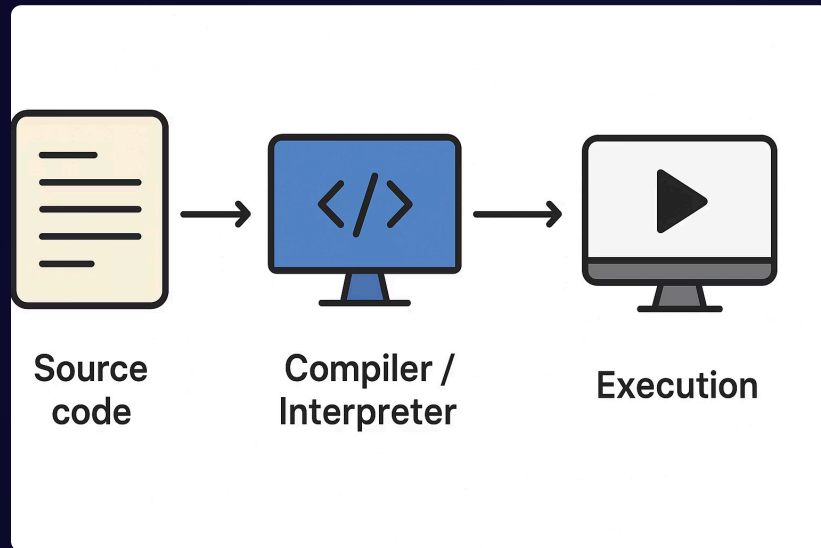


SHELL : CLI program to the OS (manage files, make files etc.): Bash, Zsh, Powershell (Windows).

Run the SHELL on app called Terminal.

The session on September 24th goes into the details of Bash.

What is source code?



Compiler/Interpreter

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They **understand specific instructions** written in a programming language and convert them into something the machine can execute. e.g., Python (Interpreter), C++ (Compiler), etc.

A compiler **translates the source code into another form** (usually machine code or bytecode) for execution. An interpreter, in contrast, directly executes instructions line by line.

Source Code

T

The recipe (written in text, as a text file) for the specific compiler/interpreter.

File extensions like `.py` or `.cpp` are mainly for human readability, editors, and tooling; essentially, they are all text files.

	Code editors	IDEs
 Feature set	Provides standard code editing functionality, with modern code editors including more advanced functionality.	Provides comprehensive functionality.
 Language support	Polyglot with support for additional languages through plugins or extensions.	Provides extensive support for specific programming languages and frameworks.
 Extensibility	Lightweight feature set that can be built on with community-driven extensions.	Large feature set already built in with APIs for developing community extensions.
 Resource usage	Consumes fewer system resources.	Consumes more system resources.
 Learning curve	Generally simpler, less complex, and easier to learn.	Generally a wider range of features, more complex, and harder to learn.
 Integration with external tools	Integrates with external tools through plugins or extensions, which can need some configuration.	Preconfigured integration with external tools and services commonly used in software development.
 Examples	VS Code, Cursor, Zed, Nova, Sublime Text**, Vim**, Notepad++** **Text editors	JetBrains IDEs (IntelliJ IDEA, WebStorm, PhpStorm, PyCharm, Rider), Visual Studio, Eclipse

VSCode

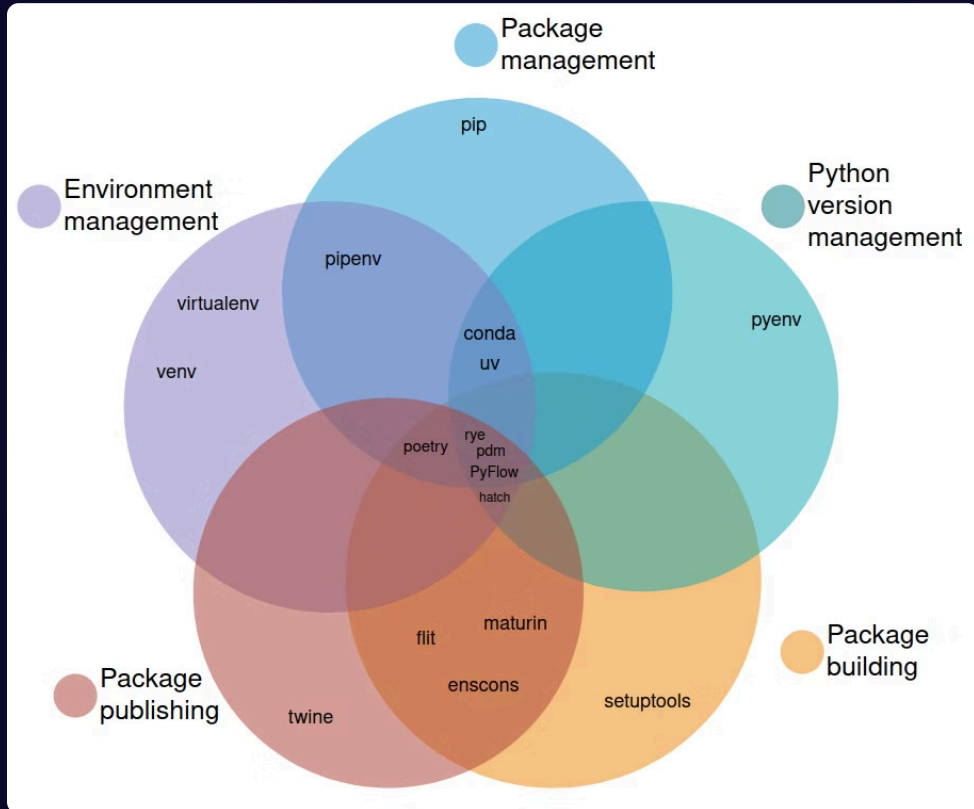


Light-weight code editor that can be configured to be an IDE for any programming language.

Extensions for AI coding agents.

[Download VSCode](#)

Python Environment Management



Popular choices:



`pip + venv + pyenv`



`conda` (Anaconda, Miniconda)



`uv` (fast, very flexible)

More detailed discussion during the session on 5th & 6th February.

Git

The Git logo, which consists of the word "git" in a lowercase, sans-serif font, enclosed within a dark blue circle.

Version Control

Record and track all changes to all files in a directory (repository).



Branches

Create branches (or copies) for parallel development on different parts.



Locally

Version controlling locally on your machine.



GitHub hosting

Host your Git repositories on cloud using GitHub.



More on January 14

My Project Structure

Each project in a separate directory.



Separate venv for each project.

VSCode



Version controlling locally on your machine.



UV

Host your Git repositories on cloud using GitHub.

Ask me anything!

Thank you and Good Luck!